

OE-EC803A — Internet of Things

Semester VIII • 3L:0T:0P • 3 Credits

Module 4: Prototyping Physical Design & Enclosures

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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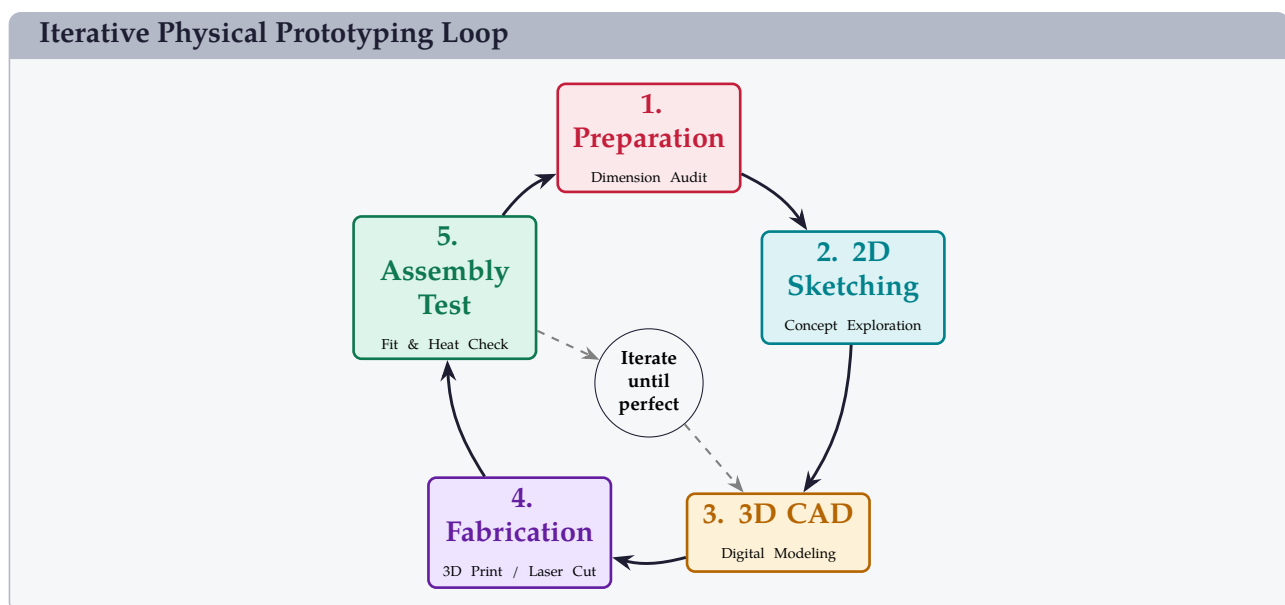
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1. The Physical Prototyping Process

Physical design defines the mechanical structure, look, and user feel of an IoT device. An enclosure must protect the internal electronics while providing access to pins, ports, and sensors.

1.1 Prototyping Workflow Steps

- **Preparation:** Gathering circuit dimensions, mounting hole coordinates, and connectors.
- **Sketching & Exploration:** Creating rapid 2D hand sketches to study button placements and accessibility.
- **Iteration:** Generating digital CAD models and manufacturing initial prototypes.
- **Testing:** Assembling the electronics inside the case to check for connector alignment, heat dissipation, and physical tolerances.



2. Prototyping Methods

Various low-fidelity and high-fidelity fabrication methods are used depending on the design stage.

2.1 Non-Digital Prototyping (Low-Fidelity)

Before using digital machines, designers build fast, manual models using cardboard, foam core, acrylic sheets, and modeling clay.

- **Advantages:** Extremely low cost, zero machine wait time, and allows testing basic dimensions in a few minutes.
- **Drawback:** Low precision, fragile, and cannot be used for mechanical testing.

2.2 Laser Cutting (2D Vector Fabrication)

Laser cutting uses a high-power laser beam to cut or engrave sheet materials (acrylic, MDF, plywood, cardboard) based on 2D vector drawing profiles.

- **Vector Layouts:** Drawn in programs like Inkscape or AutoCAD using closed paths.
- **Kerf:** The width of the material that is vaporized by the laser beam during cutting (typically 0.1mm to 0.3mm). To create tight press-fit or finger joints, the design dimensions must compensate for this kerf allowance.
- **Joint Assembly Types:**
 - *Finger/Box Joints:* Interlocking rectangular teeth along edges, glued or press-fit.
 - *T-Slot Joints:* A slot receives a nut, and a perpendicular screw pulls the interlocking face tight. Excellent for structures needing frequent disassembly.
 - *Mortise and Tenon:* A tab (tenon) fits tightly into a matching slot (mortise) to align panels.

2.3 3D Printing (Additive Manufacturing)

3D Printing (Additive)

3D Printing is an additive manufacturing process that builds 3D solid structures layer-by-layer from a digital CAD file. The dominant technology is **FDM (Fused Deposition Modeling)**, which extrudes molten thermoplastic filament (PLA, ABS) through a heated nozzle.

Key 3D printing parameters that affect enclosure durability and print time:

- **Layer Height:** Controls vertical resolution (typically 0.1mm to 0.3mm). Smaller values give smoother surfaces but increase print time.
- **Infill Density & Patterns:** The internal grid structure, ranging from 0% (hollow) to 100% (solid). Enclosures typically use 15% to 30% infill. Patterns include *Honeycomb* (strongest in multiple axes), *Gyroid* (isometric strength and flexibility), and *Grid* (fastest printing).
- **Wall/Shell Thickness:** The thickness of the outer boundaries, defining enclosure strength and screw-mounting thread durability.
- **Materials:**
 - *PLA (Polylactic Acid):* Biodegradable starch-based polymer. Easy to print, low warping, but low temperature resistance (warps above 60°C).
 - *ABS (Acrylonitrile Butadiene Styrene):* Petroleum-based. Highly impact-resistant, high heat deflection, but requires a heated enclosure and emits toxic fumes during printing.
 - *PETG (Polyethylene Terephthalate Glycol):* Blends ease of PLA with durability of ABS. Excellent chemical and weather resistance, ideal for outdoor IoT nodes.

2.4 CNC Milling (Subtractive Manufacturing)

CNC (Computer Numerical Control) milling uses rotating cutting bits to carve away material from a solid block of plastic, wood, or metal (aluminum).

- **Subtractive Concept:** The opposite of 3D printing. It starts with a solid stock block and subtracts material using high-speed rotary cutters.
- **Milling Toolpaths:** G-code instructions define specific cutter movements:
 - *Pocketing:* Carving out internal hollow cavities inside a solid block (e.g. creating the box cavity for electronics).
 - *Contouring (Profiling):* Cutting along outer boundary lines to separate the final piece from raw stock.
 - *Facing:* Machining flat surfaces on a block to ensure absolute thickness tolerance.
- **Applications:** Best for high-strength enclosures, metal heat-sink cases, and prototyping circuit boards (PCB isolation routing).

2.5 Repurposing and Recycling

Hacking off-the-shelf plastic junction boxes, containers, or toys by drilling holes for cables and sensors.

- **Advantages:** Provides immediate structural protection, is highly cost-effective, and reduces e-waste footprint during research stages.

Quick Revision — Module 4

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One-Line Definitions

- **Kerf:** The thickness of the cut line or material slot vaporized by a laser cutter.
- **FDM:** A 3D printing process that builds objects by extruding layers of melted plastic filament.
- **Subtractive Manufacturing:** A fabrication process where material is carved away from a solid block.
- **Additive Manufacturing:** A fabrication process where material is joined layer-by-layer to build a structure.
- **Infill:** The internal supporting pattern printed inside a 3D object to save time and material.

Understanding the kerf is critical for press-fit joints where interlocking teeth must hold together without glue.

Frequently Asked / Viva Questions

Q1. What is the difference between additive and subtractive manufacturing?

Additive manufacturing (e.g. 3D printing) builds parts layer-by-layer from nothing, producing zero waste and allowing complex internal cavities. Subtractive manufacturing (e.g. CNC milling) carves parts out of a solid block of raw material, producing waste chips but yielding high strength and support for metals.

Q2. Explain the significance of “kerf” in laser cutting.

The kerf is the width of the cut slot created by the laser beam vaporizing material. If a laser has a kerf of 0.2mm, cutting a 10.0mm square will result in a piece that is 9.8mm wide. Designers must compensate for this kerf to create tight, interlocking press-fit joints.

Q3. Why is infill density in 3D printing rarely set to 100%?

A 100% infill uses massive amounts of filament and takes a very long time to print. Structural strength gains level off significantly beyond 40% infill. Enclosures are usually printed with 15% to 30% infill using internal patterns (like grid or gyroid) to maximize strength-to-weight ratio.

Q4. What is Fused Deposition Modeling (FDM)?

FDM is the most common 3D printing technology. It melts a plastic filament (like PLA or ABS) and extrudes it through a small nozzle, moving in X, Y, and Z axes to lay down thin layers that fuse together as they cool.

Frequently Asked / Viva Questions (contd.)

Q5. Why would a designer choose CNC milling over 3D printing for an industrial IoT gateway enclosure?

CNC milling allows using materials like structural aluminum. Aluminum enclosures provide superior mechanical protection, complete electromagnetic shielding (Faraday cage), and act as built-in heat sinks for high-power processors.

Q6. What is the purpose of low-fidelity physical prototyping using cardboard?

Low-fidelity prototyping allows designers to physically hold and check the dimensions of a device layout within minutes, helping catch major interface errors (like port placement blockage or button hand-reach issues) before executing expensive machine runs.

Q7. What is repurposing/recycling in physical design?

Repurposing refers to taking an existing mass-produced plastic box (like an electrical junction box or soap container) and drilling custom ports to house the IoT circuit. This saves cost, is weather-sealed, and reduces development time.

Q8. What is SLA (Stereolithography) 3D printing, and how does it differ from FDM?

SLA uses a liquid photopolymer resin that is cured layer-by-layer using a UV laser beam. It yields extremely high resolutions and smooth surface finishes compared to FDM, but the materials are more fragile and require post-print chemical cleaning.

Comparison Snapshot

3D Printing vs. CNC Milling vs. Laser Cutting:

Feature	3D Printing (FDM)	CNC Milling	Laser Cutting
Process	Additive (layer-by-layer).	Subtractive (carving).	2D Cutting / Engraving.
Dimensions	Complex 3D geometries.	3D shapes with tool limitations.	Flat 2D sheets (can fold/slot).
Materials	PLA, ABS, PETG.	Metals (Al, Steel), Wood, Acetal.	Acrylic, MDF, Plywood, Cardboard.
Wastage	Very Low (only support material).	High (residual chips and dust).	Low-Medium (sheet offcuts).
Setup Speed	Very Fast (load STL to printer).	Slow (requires CAM setups, tools).	Fast (load vector file).